HEXADECIMAL & COLOUR CODES

COLOURS IN THE DIGITAL WORLD

EVERYTHING YOU SEE ON SCREENS IS MADE OF COLORS!

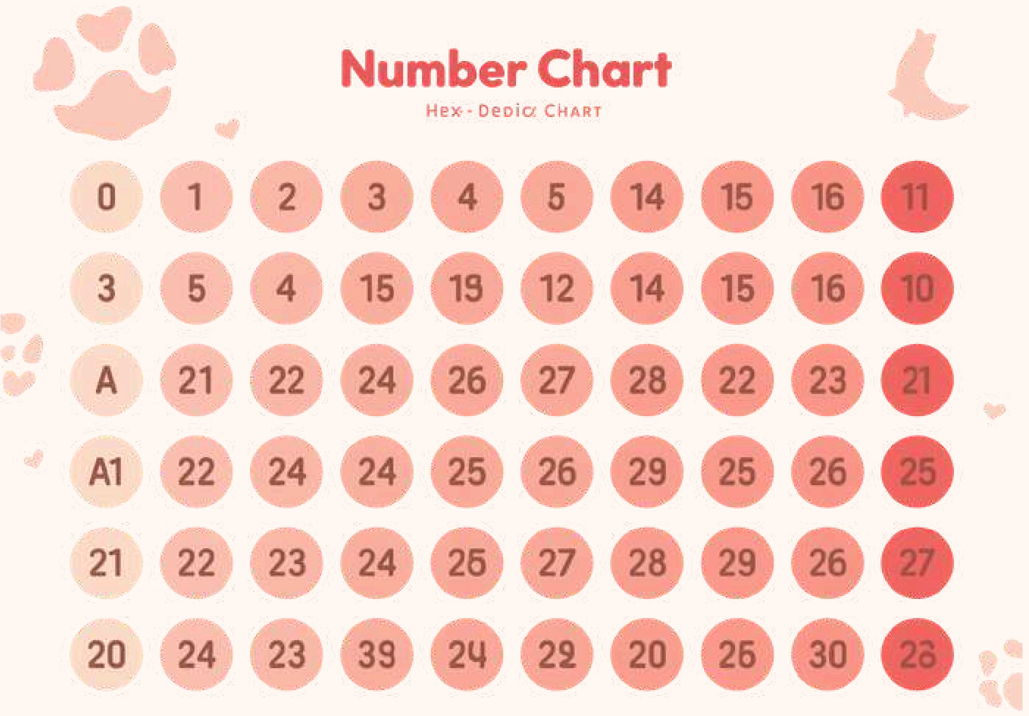
Screens are made of tiny dots called pixels. Each pixel glows with digital colors made from light. Understanding colors helps us create amazing things with computers and AI!

WHAT ARE HEXADECIMAL NUMBERS?

Hexadecimal is a special way to count using 16 digits instead of just 10!

Regular counting: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9

Hexadecimal adds: A, B, C, D, E, F



LIKE COUNTING BEYOND 9!

Think of it like this: when we run out of single digits at 9, we add letters to keep counting!

A=10, B=11, C=12, D=13, E=14, F=15

THE # SYMBOL IN COLOUR CODES

The # symbol is like a special signal that tells computers "this is a color code!" Whenever you see # followed by letters and numbers, you know it's describing a specific color.

Example: #FF5733
This code creates a beautiful orange-red colour!

UNDERSTANDING HEX COLOUR CODES

1. RED (FIRST 2 DIGITS)

The first pair controls how much red light appears in your colour!

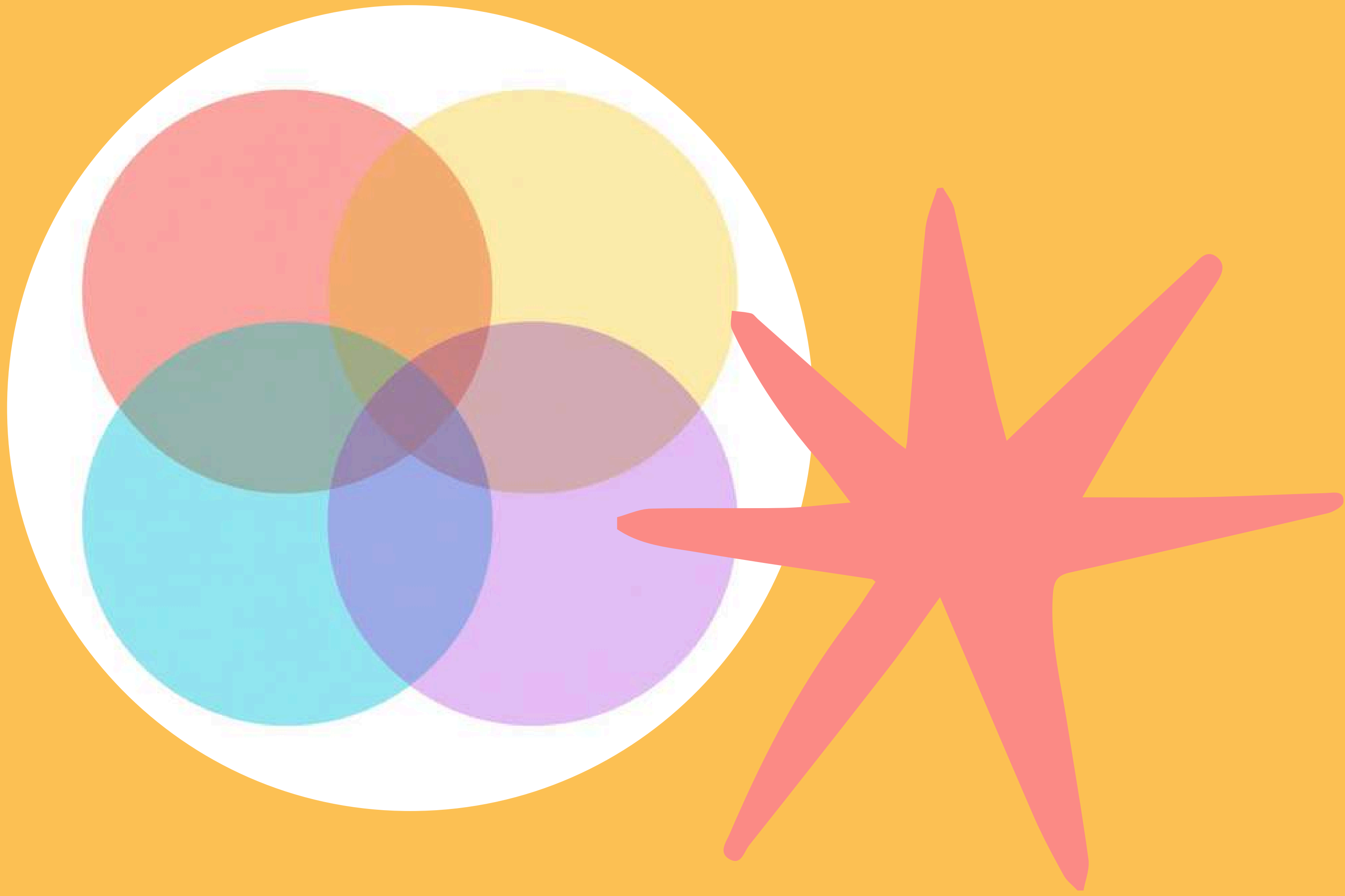
1. GREEN (MIDDLE 2 DIGITS)

The middle pair sets how much green light blends into your colour!

1. BLUE (LAST 2 DIGITS)

The last pair decides the blue light intensity in the mix!

THE RGB COLOUR MODEL



Red, Green, and Blue light combine to make all colors! These three primary colors of light mix together in different amounts to create every color you see on screens, from bright yellows to deep purples.

BREAKING DOWN #FF5733

#FF5733 breaks down into three parts:

- Red = FF (maximum intensity!)
- Green = 57 (medium intensity)
- Blue = 33 (lower intensity)

This creates a beautiful coral-orange color!

HEX VALUES EXPLAINED

Hex values range from 00 (none) to FF (full). Think of it like a volume dial - 00 is silent, FF is the loudest!

PRIMARY COLOURS IN HEX

1. RED

#FF0000

Pure red light at maximum brightness!

1. GREEN

#00FF00

Pure green light at full power!

1. BLUE

#0000FF

Pure blue light shining bright!

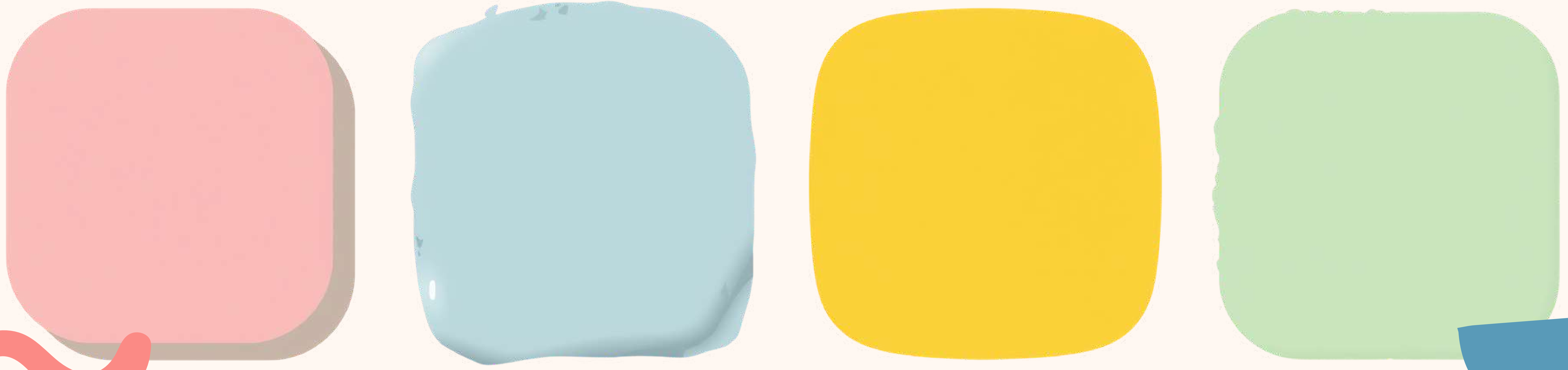
CREATING DIFFERENT COLOURS

#FF0000 (Bright Red)
Full red, no green
No blue at all

#FFAA00 (Orange)
Adding green to red
Creates a warm orange

FUN HEX COLOUR EXAMPLES

Playful hex colours kids might like! Each colour has a special code that computers understand.



DESIGNING A DIGITAL RAINBOW

STEP-BY-STEP GUIDE

1. Start with Red (#FF0000)
2. Add Orange (#FF7F00)
3. Then Yellow (#FFFF00)
4. Next Green (#00FF00)
5. Add Blue (#0000FF)
6. Finish with Violet (#8B00FF)

WATCH THE RAINBOW APPEAR!

By gradually changing hex values, you create a beautiful spectrum of colors on your screen!

ACTIVITY TIME! GUESS THE COLOUR

#00FF7F

Can you guess what colours these are?

#FF69B4

#1E90FF

Check with a color picker!

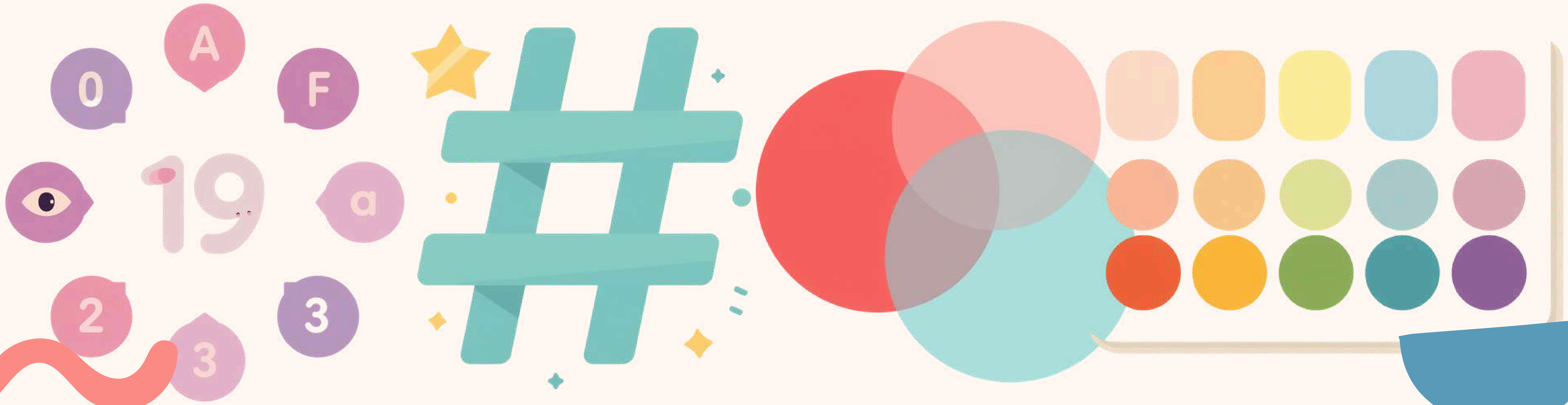
REAL-WORLD APPLICATIONS OF HEX COLOURS

FROM WEBSITES TO VIDEO GAMES, HEX COLOURS HELP DESIGNERS!

Hex colour codes are used everywhere in the digital world! Website designers use them to pick perfect colours for buttons and backgrounds. Game developers use hex codes to create colourful characters and worlds. App creators choose hex colours to make their apps look beautiful. Even digital artists use hex codes to paint amazing artwork on computers!

RECAP & WHAT YOU LEARNED

Great job learning about hex colours! Here are the key points to remember from our colourful journey today.



THANK YOU & KEEP EXPLORING COLOURS!

Thanks for learning about hex colors! Keep exploring and creating!